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Ahmedabad University

CSE 516 – Probabilistic Graphical Models

Milestone 3: Results and Analysis

Bayesian Personalized Recommendation using Latent Trait Models & Expectation Propagation

Category: Theoretical AI

Group 4

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GitHub Repository

https://github.com/Jyotier2006/M2_g4_th_apl_recommend-sys_team

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Abstract

We extend the existing M2 study of a restricted static Matchbox recommender with systematic full-data experiments and an explicit Gaussian Markov movie prior. Across five MovieLens 1M splits, ADF, EP5 and EP10 achieve expected-rating RMSE 0.8977 ± 0.0035, 0.8643 ± 0.0037 and 0.8548 ± 0.0029. GMRF-MAP achieves 0.8499 ± 0.0023, but its matched graph-free MAP baseline already reaches 0.8506 ± 0.0021: the isolated graph gain is modest. Paired user-bootstrap intervals support a reduction in four of five splits; one includes zero. Repeated EP improves NLL while worsening ECE, and changing observation noise does not uniformly improve probability quality. Missing-data, corrupted-label, graph-density, metadata and cold-start tests probe robustness. Exact Gaussian and numerical quadrature diagnostics expose what the approximations preserve and lose. These results position the project as an inference and regularization study, not a full Matchbox/PreRec reproduction or a state-of-the-art claim.

2 Problem and cumulative progress from M1/M2

Sparse user–item ratings leave uncertainty about both preferences and item characteristics. A latent-trait graphical model shares evidence across ratings through common user and item parameters, while Bayesian learning retains uncertainty about those parameters. Our original theoretical question remains: *how do ADF and repeated EP differ in predictive quality, uncertainty, computation and memory?* M3 adds: *does an explicit movie Markov prior improve a matching matrix-factorization model, and under which conditions?*

This is an update of the M2 report. The project identity, latent-trait model, EP/VMP equations and validation foundation are retained. M3 expands the benchmark to five seeds, adds MAP and graph ablations, studies both assumed noise and corrupted observations, and replaces the preliminary conclusion with evidence from measured runs. The earlier synthetic offline demo is not used as MovieLens evidence. The original M2 result files remain separately labelled for provenance.

The work retains the Theoretical AI track with recommendation as its benchmark. The new graph is constructed from training similarities, not learned through causal structure discovery. User-specific thresholds and temporal dynamics remain outside the implemented scope. We compare several meaningful approaches; we do not claim to test every possible PGM algorithm.

3 Probabilistic representation

For user u and item i, let $s_u, t_i \in \mathbb{R}^K$ denote latent traits and b_u, b_i denote biases. For each observed pair $(u, i) \in \mathcal{D}$,

$$z_{uik} = s_{uk} t_{ik}, \qquad a_{ui} = \mu + b_u + b_i + \sum_{k=1}^{K} z_{uik}, \tag{1}$$

$$y_{ui} \mid a_{ui} \sim \mathcal{N}(a_{ui}, \beta^2), \qquad R_{ui} = r \iff c_{r-1} < y_{ui} \le c_r. \tag{2}$$

Only the rating is observed. The cuts are $c = (-\infty, 1.5, 2.5, 3.5, 4.5, +\infty)$; μ is the training mean, treated as fixed. Independent priors are

$$s_{uk} \sim \mathcal{N}(m^0_{uk}, 0.25), \quad t_{ik} \sim \mathcal{N}(n^0_{ik}, 0.25), \quad b_u, b_i \sim \mathcal{N}(0, 1). \tag{3}$$

Trait prior means are sampled once from $\mathcal{N}(0, 0.1^2)$ using the run seed. This breaks bilinear sign symmetry without using held-out ratings.

With $\Theta = (S, T, b_U, b_I)$, the joint factorization is

$$\begin{aligned} p(\Theta, Z, A, Y, R) = p(\Theta) \prod_{(u,i)\in\mathcal{D}} \Bigg[\prod_k \delta(z_{uik} - s_{uk} t_{ik}) \Bigg] \\ \times \delta\Bigg(a_{ui} - \mu - b_u - b_i - \sum_k z_{uik} \Bigg) \\ \times \mathcal{N}(y_{ui}; a_{ui}, \beta^2) \mathbf{1}\{c_{R_{ui}-1} < y_{ui} \le c_{R_{ui}}\}. \end{aligned} \tag{4}$$

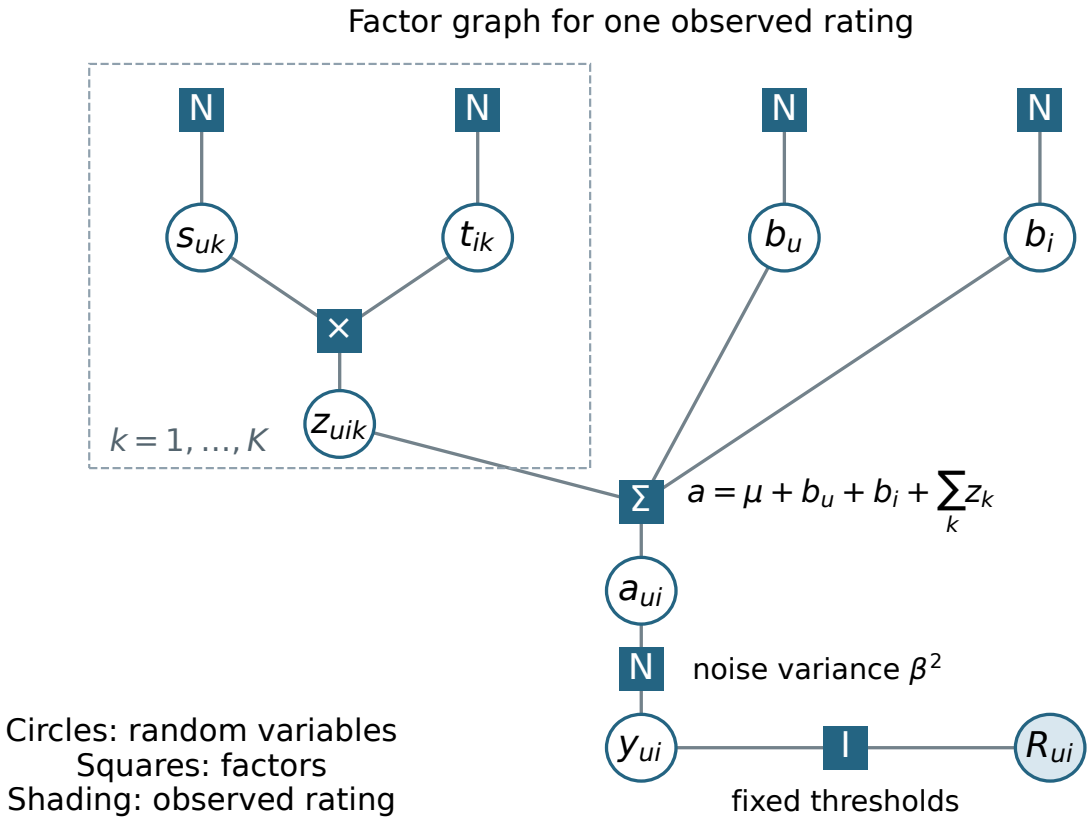


Figure 1: Factor graph for one rating. The plate repeats the trait product; user/item variables are shared across all corresponding ratings. Edges connect variables to the factors that depend on them.

Prior factors regularize parameters; products express compatibility; the sum adds biases; Gaussian noise captures residual variation; and the interval factor converts a continuous score into a rating. The preserved M2 reference includes a concrete variable/factor/edge specification in `graph_instance.json`.

Ratings are conditionally independent given all relevant parameters. Ratings sharing a user or item are generally dependent after marginalization. Independent priors do not imply independent posterior parameters: the factorized Gaussian posterior is an approximation. The model assumes stationary preferences and does not model why a movie was rated or identify causal effects.

4 Learning and inference

4.1 Posterior learning and prediction query

Training approximates $p(\Theta \mid R_{\mathcal{D}})$ by a product of univariate Gaussians, storing precision $\lambda = 1/v$ and weighted mean $\eta = m/v$. Thresholds remain fixed; noise and prior variances are fixed within a run and varied in the M3 sensitivity experiments. Prediction computes $p(R_{ui} = r \mid R_{\mathcal{D}})$ for an unobserved pair.

Under the factorized approximation, product moments are

$$M_k = \mathbb{E}[s_k t_k] = m_{s_k} m_{t_k}, \tag{5}$$

$$V_k = \mathrm{Var}(s_k t_k) = v_{s_k} v_{t_k} + v_{s_k} m_{t_k}^2 + v_{t_k} m_{s_k}^2. \tag{6}$$

Set $m_y = \mu + m_{b_u} + m_{b_i} + \sum_k M_k$ and $v_y = \beta^2 + v_{b_u} + v_{b_i} + \sum_k V_k$. Approximating the score as Gaussian gives

$$\widehat{p}_r = \Phi\left(\frac{c_r - m_y}{\sqrt{v_y}}\right) - \Phi\left(\frac{c_{r-1} - m_y}{\sqrt{v_y}}\right), \qquad \widehat{R} = \sum_{r=1}^{5} r\widehat{p}_r. \tag{7}$$

The moments are exact under the factorized q; the Gaussian score approximation is additional. Probabilities use a 10^{-15} numerical floor and renormalization.

4.2 Local hybrid EP/VMP update

Exact posterior inference is intractable at this scale because bilinear products and ordinal observations couple many parameters. Following Matchbox [4], we use VMP at products and EP or exact Gaussian messages elsewhere. A local loop alternates forward moments, ordinal moment matching and reverse messages, reconstructing marginals from a fixed cavity and the current message.

For an incoming score $\mathcal{N}(m, v)$, define $\alpha = (c_{r-1} - m)/\sqrt{v}$, $\gamma = (c_r - m)/\sqrt{v}$ and $Z = \Phi(\gamma) - \Phi(\alpha)$. Truncated moments are

$$\widehat{m} = m + \sqrt{v}\,\frac{\phi(\alpha) - \phi(\gamma)}{Z}, \tag{8}$$

$$\widehat{v} = v\left[1 + \frac{\alpha\phi(\alpha) - \gamma\phi(\gamma)}{Z} - \left(\frac{\phi(\alpha) - \phi(\gamma)}{Z}\right)^2\right]. \tag{9}$$

Gaussian division yields $\lambda_o = \widehat{v}^{-1} - v^{-1}$ and $\eta_o = \widehat{m}/\widehat{v} - m/v$. Write the outgoing Gaussian as $\mathcal{N}(y; m_o, v_o)$. Reverse propagation to product z_k gives

$$d_k = m_o - \mu - m_{b_u}^{\setminus n} - m_{b_i}^{\setminus n} - \sum_{\ell \neq k} M_\ell, \tag{10}$$

$$w_k = v_o + \beta^2 + v_{b_u}^{\setminus n} + v_{b_i}^{\setminus n} + \sum_{\ell \neq k} V_\ell. \tag{11}$$

The VMP message to s_k is

$$\lambda_{f \to s_k} = \frac{\mathbb{E}[t_k^2]}{w_k}, \qquad \eta_{f \to s_k} = \frac{d_k \mathbb{E}[t_k]}{w_k}, \quad \mathbb{E}[t_k^2] = v_{t_k} + m_{t_k}^2. \tag{12}$$

The item update is symmetric. A bias receives the analogous Gaussian sum message with its own cavity term omitted. Product moments use current local marginals; bias inputs remain cavities. Local damping is 0.7, with at most 10 iterations and an absolute mean/variance change tolerance of 10^{-4}.

4.3 ADF versus repeated EP

EP caches a Gaussian site per rating. For each touched scalar parameter,

$$\lambda = \lambda_0 + \sum_n \lambda_n, \quad \eta = \eta_0 + \sum_n \eta_n; \qquad \lambda^{\setminus n} = \lambda - \lambda_n, \quad \eta^{\setminus n} = \eta - \eta_n. \tag{13}$$

An EP pass visits the fixed training order, subtracts the old site's natural parameters, runs the local update, and replaces that site. The replacement is damped with weight 1 on pass one and 0.5 thereafter.

ADF processes each rating once and discards its site. Thus EP's first pass matches ADF under identical priors and order. Replaying ADF without cavity removal would double-count evidence. Our ADF interface rejects a second pass. Five or ten EP passes are computational budgets, not proof of convergence.

5 M3 extension: an explicit Markov movie network

5.1 What changed, and why it might help

The M2 latent-trait model already is probabilistic. Its independent movie priors, however, do not explicitly connect similar movies. M3 adds a separate Gaussian Markov random field (GMRF) prior to a matrix-factorization model. An undirected edge encourages two movies' latent vectors to remain similar. This can reduce estimation variance when ratings are sparse or unreliable, but a misleading or overly strong edge can introduce bias. Improvement is a hypothesis to test, not a property guaranteed by the word "Markov."

The new model uses maximum a posteriori (MAP) estimation by alternating block minimization. It does not add graph messages to the original EP implementation. Consequently, comparing GMRF-MAP with EP changes both the prior and the fitting method/likelihood. The controlled graph ablation is **GMRF-MAP versus MF-MAP with** $\gamma = 0$, sharing the data, initialization, regularization, training budget and validation rules.

5.2 Building the graph from available evidence

For each split, the graph is constructed from training ratings and fixed movie genres only. Five alternating regularized bias updates estimate μ, b_u, b_i; their residuals are $e_{ui} = R_{ui} - \mu - b_u - b_i$. Each movie's sparse residual column is normalized. Positive residual cosine similarities are multiplied by $n_{ij}/(n_{ij} + 20)$, where n_{ij} is the number of common training users; similarities based on fewer than five common users are set to zero.

Genres are represented by an 18-dimensional multi-hot vector, normalized to unit length before computing genre cosine similarity. Default affinities mix 90% residual similarity and 10% genre similarity. Each movie retains its top 20 positive neighbours; ties use mapped movie IDs. The union is symmetrized using the maximum of the two directed weights, so final degrees can exceed 20. If A denotes this symmetric affinity, $W = D_A^{-1/2} A D_A^{-1/2}$. We then recompute $D_{ii} = \sum_j W_{ij}$ and set $L = D - W$. The diagonal of W is zero. This is a deterministic similarity construction, not causal discovery or recovery of a known true graph. Across the five main splits, the graph has 56,408–56,575 undirected movie edges (mean 56,508), mean degree 30.50, density 0.823%, and no isolated movie nodes. These are representation statistics, not edge-recovery accuracy. Because W is estimated from training ratings, this

is a data-dependent, empirical-Bayes-style regularization construction. We condition on the resulting W and do not propagate uncertainty about its edges.

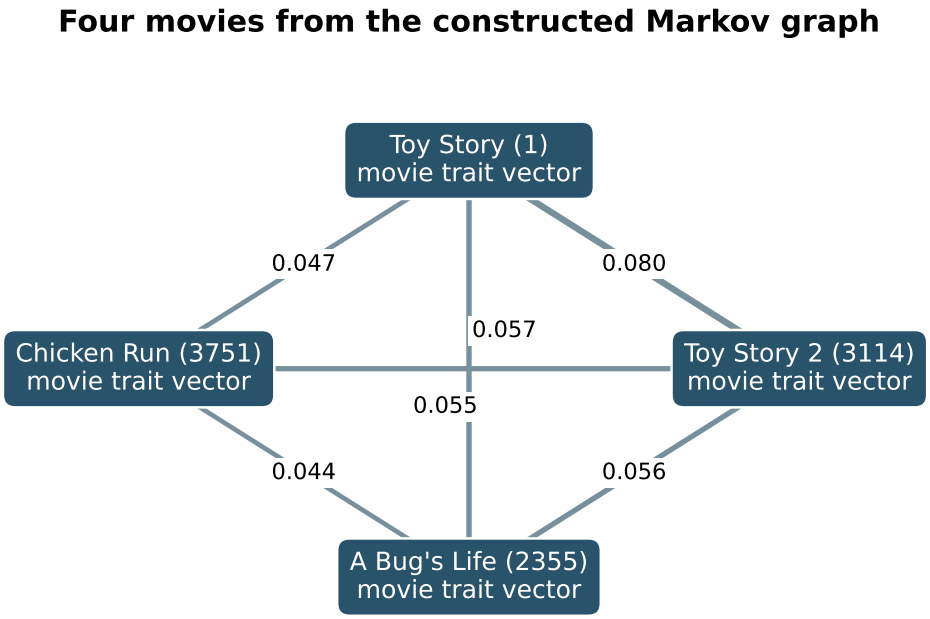


Figure 2: Actual seed-42 subgraph: Toy Story and its three strongest neighbours. Node labels give titles and raw MovieLens IDs; displayed edge labels are the normalized weights. Each node represents a movie's latent vector; each edge induces the pairwise factor in Equation 14. The full graph has many other neighbours. Similarity is not a causal claim.

5.3 Factors, conditional independence and MAP objective

With $Q_i \in \mathbb{R}^K$ for movie i, the movie prior is

$$p(Q \mid W) \propto \prod_i \exp\left(-\frac{\lambda}{2}\|Q_i\|^2\right) \prod_{\{i,j\}\in E} \exp\left(-\frac{\gamma w_{ij}}{2}\|Q_i - Q_j\|^2\right), \tag{14}$$

$$J = \lambda I + \gamma L, \qquad \operatorname{tr}(Q^T L Q) = \sum_{\{i,j\}\in E} w_{ij}\|Q_i - Q_j\|^2. \tag{15}$$

Each trait coordinate across movies has Gaussian precision J. For any nonzero x, $x^T J x = \lambda\|x\|^2 + \gamma \sum_{\{i,j\}\in E} w_{ij}(x_i - x_j)^2 > 0$ when $\lambda > 0$. Thus the prior is proper, including for disconnected graphs. A zero off-diagonal entry in J expresses conditional independence in this prior, given all other movie variables. It does not imply marginal independence or the absence of posterior dependence after observing ratings.

The conditional for one coordinate of movie i is

$$Q_{ik} \mid Q_{-i,k}, W \sim \mathcal{N}\left(\frac{\gamma \sum_j w_{ij} Q_{jk}}{\lambda + \gamma D_{ii}}, \frac{1}{\lambda + \gamma D_{ii}}\right). \tag{16}$$

This gives a precise meaning to "neighbours share information." Nodes are latent variables, and edges encode pairwise probability factors; they are not causal arrows, a neural network, or time transitions in a Markov chain.

Using $f_{ui} = \mu + b_u + b_i + P_u^T Q_i$, independent Gaussian user priors, bias priors and unit-variance Gaussian numeric-rating likelihood gives the energy

$$\begin{aligned}\mathcal{E} = \frac{1}{2} \sum_{(u,i)\in\mathcal{D}} (R_{ui} - f_{ui})^2 + \frac{\lambda}{2}(\|P\|_F^2 + \|Q\|_F^2) \\ + \frac{\lambda_b}{2}(\|b_U\|^2 + \|b_I\|^2) + \frac{\gamma}{2}\operatorname{tr}(Q^T L Q).\end{aligned} \tag{17}$$

Defaults are $K = 8$ and $\lambda = \lambda_b = 10$. Holding other blocks fixed, each user/movie vector and its bias solve a $(K+1)$-dimensional linear system. For a movie, the trait block gains $\gamma D_{ii} I$ on the left and $\gamma \sum_j w_{ij} Q_j$ on the right. Sequential movie updates use the newest neighbours (Gauss–Seidel). Each exact block update cannot increase this energy; joint bilinear optimization remains nonconvex, so this is not a global-optimum guarantee. Source: `gmrf.py`, functions `graph`, `als` and `objective`.

5.4 Prediction and the meaning of uncertainty

MAP stores point estimates of the traits and biases. A single Gaussian observation variance is fitted on validation NLL and integrated between the same five rating intervals as EP. This produces probabilities and an expected star rating for common KPI evaluation. The separately reported clipped score $\min(5, \max(1, f_{ui}))$ is another point decoder. The MAP probability wrapper represents residual variation; it is **not a full Bayesian posterior** over the fitted parameters. EP retains approximate parameter uncertainty but uses a factorized posterior and additional score-Gaussian approximation.

For T epochs, dense small-block solves cost $O(T[N(K+1)^2 + (U+I)(K+1)^3 + |E|K])$. The sparse graph requires $O(|E|)$ storage. Graph construction evaluates movie similarities in row blocks and can be costly for a large catalogue; batching reduces peak dense memory, not its pairwise nature. Graph building and hyperparameter search are reported in addition to fitting the selected candidate.

6 Mathematical validation and complexity

The M2 mathematical foundation retained nine checks: truncated moments against quadrature; exact marginals for a single $K = 0$ ordinal factor; ADF–EP1 equality; the prior-plus-sites invariant; no double counting on a single-rating revisit; product moments against Monte Carlo; probability/metric sanity; disjoint reproducible splits; and checkpoint reload. These validate specific formulas and invariants, not exactness of the whole approximation.

The bilinear test, rerun for M3, uses $R = 5$, offset 3, noise variance 1, $s \sim \mathcal{N}(0.5, 0.4)$ and $t \sim \mathcal{N}(0.8, 0.4)$, with bias variances 10^{-8}. Tensor Gauss–Hermite quadrature integrates the likelihood $P(y > 4.5 \mid s, t)$, analytically integrating the biases. Increasing the order from 40 to 80 changes the reference moments by at most 1.04×10^{-10}. For this diagnostic only, the local cap is 500 and tolerance 10^{-12}; it converges after 50 iterations.

Means differ by less than 0.013, but both variances are about 10% too small. This motivates checking uncertainty as well as rating error.

For N ratings, K traits, L local iterations and T passes, training costs $O(TNL(K+1))$ and batch prediction $O(N(K+1))$. Posterior storage is $O((U+I)(K+1))$. Repeated EP additionally stores

$$8 \times 2 \times N(2K+2) = 32N(K+1) \text{ bytes} \tag{18}$$

Quantity	Quadrature	Hybrid	Absolute error
$\mathbb{E}[s]$	0.922831	0.933888	0.011056
$\mathrm{Var}(s)$	0.325234	0.292029	0.033206
$\mathbb{E}[t]$	1.123960	1.136053	0.012093
$\mathrm{Var}(t)$	0.351319	0.315457	0.035862
$\mathrm{Cov}(s,t)$	0.070526	0.000000	0.070526

Table 1: One-observation validation against 80-point-per-axis quadrature. The hybrid approximation cannot retain the nonzero posterior covariance.

in 64-bit rating sites. At $N = 800{,}167$, $K = 8$, this is 219.77 MiB; ADF has no site cache. The 87,714 scalar parameters require 175,428 natural parameters (1.34 MiB). These array sizes exclude data, temporaries and process overhead.

7 Updated benchmark, baselines and experimental protocol

7.1 Dataset and code representation

MovieLens 1M [1, 2] contains 1,000,209 ratings, 6,040 users and 3,706 rated movies; its metadata catalogue has 3,883 movies. SHA-256 checks verify the official archive and all three input files. Rating density is 4.468%; absent user–movie pairs are unobserved, never zero-star labels.

File / field	Meaning	Representation / use
ratings: UserID, MovieID	Raw entity identifiers	Zero-based arrays `u`, `i`; mappings `uid`, `iid`
ratings: Rating	Observed integer, 1–5	Target array `r`; train/validation/test separated
ratings: Timestamp	Unix observation time	Audited; no temporal predictor or chronological claim
movies: MovieID, Title	Identifier and display name	ID joins; titles are not numeric features
movies: Genres	Pipe-separated genre labels	18-column multi-hot matrix used only for graph construction
users: UserID	User identifier	Join/audit only beyond the rating ID
users: Gender, Age	Recorded demographic category, age bin	Parsed/audited; excluded from prediction
users: Occupation, Zip-code	Occupation category and postal code	Parsed/audited; excluded from prediction

Table 2: All dataset fields and their actual uses. IDs are categories, not numerical magnitudes.

At $K = 8$, each user and each movie has eight unnamed learned traits plus one bias: $(6040 + 3706)(8 + 1) = 87,714$ persistent scalar latent parameters. The eight coordinates are not predefined genres. EP stores two natural parameters per scalar plus per-rating sites. GMRF adds fixed graph weights; the count above excludes observations, deterministic auxiliary variables, hyperparameters, and EP site-cache entries.

7.2 Common evaluation and validation-only selection

Five seeds (42–46) each define 800,167 training, 100,020 validation and 100,022 test rows through one seeded random permutation. All compared methods use the same rows within a seed. Seeds control split/order and initialization; 42 has no special statistical meaning. The first three splits reproduce the M2 protocol. Results from this fresh execution are reported together; old M2 timings are not mixed with current timings.

Baseline families. Global/user/item frequency predictors use the M2 smoothed class-histogram rules. MF-SGD retains the M2 kernel, $K = 8$, learning rate 0.007, epoch decay 0.98, regularization 0.03 and 25 epochs. MF-MAP and GMRF-MAP use the same alternating solver with and without the graph factor. The two MF optimizers use different objectives/regularization conventions, so their numerical regularization values are not interchangeable.

EP family. ADF and EP1–5 share the original priors and hybrid local updates; EP10 studies a larger budget. EP1, EP2, etc. mean the number of passes, not different models. An additional experiment fits EP5 at observation variances $\beta^2 \in \{0.5, 1, 2\}$; validation NLL selects the noise candidate. All fixed candidates are also reported to reveal sensitivity.

Graph family. The clean-data search uses $\gamma \in \{0, 1, 10, 100\}$ and 25 epochs per candidate. Validation clipped-score RMSE selects the checkpoint and graph strength. `GMRF_MAP` reports the best positive strength as a graph diagnostic; `Graph_selected` permits $\gamma = 0$, so a deployment choice can switch the graph off. MF-MAP is the matching $\gamma = 0$ ablation. Every candidate uses the same initialization and epoch budget. The Gaussian prediction scale is then fitted on validation NLL. No test metric enters these selection functions.

7.3 Robustness, ablations and noise

Three seeds (42–44) are used for controlled stress tests, with validation and test labels kept clean. Two different meanings of noise are examined: *assumed observation variance* inside EP and *corruption of observed training labels*. They answer different questions.

Experiment	Change	Fixed comparison / interpretation
Training availability	Retain 25%, 50%, 100% of training rows	Nested seeded subsets; identical held-out rows; random missingness only
Label noise	Replace exactly 10% or 20% of training ratings by a uniformly selected different star	Same corruption for every method; graphs rebuilt from corrupted training data
Graph density	Top 5, 20 or 50 outgoing neighbours	Same mixture/regularization; symmetrized final degree can be larger
Feature ablation	Ratings only, genres only, mixed graph	Distinguishes information sources; no test-derived similarities
Entity cold-start	Separately remove all training rows of 10% of item/user identities	Evaluate only those entities' original test rows; item genres remain available
Inference sensitivity	Prior variance 0.1/0.5, damping 0.25/1, reshuffled order	EP5 against original settings, three seeds; no claim of exhaustive tuning

Table 3: Controlled experiments. Clean curves use the same three seeds as their stress conditions.

For stress tests, the graph strength is fixed at 10, the candidate retained from the clean-validation comparisons, and compared with $\gamma = 0$; checkpoints still use validation within 25

epochs. This tests robustness of that operating setting rather than repeatedly optimizing the full grid on every condition. The cold-start checkpoint criterion uses the full original validation split, while its reported test metric uses only withheld items or users. Missingness is uniform over observed training ratings, not a model of nonrandom exposure. No ratings are generated to fill missing entries.

7.4 KPIs, confidence intervals and reproducibility

For probabilities p_{nr}, $\hat{r}_n = \sum_{r=1}^{5} r p_{nr}$.

$$\text{RMSE} = \sqrt{N^{-1} \sum_n (\hat{r}_n - r_n)^2}, \qquad \text{MAE} = N^{-1} \sum_n |\hat{r}_n - r_n|, \tag{19}$$

$$\text{NLL} = -N^{-1} \sum_n \log p_{n,r_n}, \qquad \text{Brier} = N^{-1} \sum_n \sum_{r=1}^{5} (p_{nr} - \mathbf{1}\{r = r_n\})^2. \tag{20}$$

Top-class accuracy compares $\arg\max_r p_{nr}$ with the observed star, not a rounded expectation. ECE uses ten equal-width top-confidence bins and their weighted absolute confidence–accuracy gaps. Lower RMSE, MAE, NLL, Brier and ECE are better. RMSE 0.85 is an error in star units, not 85% accuracy. We also retain 90% equal-tail interval coverage/width and prediction time.

Main tables report mean ± sample SD across five seeds; robustness uses three. Since random splits overlap, we do not treat five seed scores as five independent datasets or use a naive t-test to certify significance. Instead, within each split, 2,000 paired bootstrap replicates resample users with replacement, retaining all that user's test ratings for both models. Intervals refer to the difference in rating-weighted RMSE (baseline minus comparison). They are conditional on fitted models, do not include retraining uncertainty, and do not account for dependence through shared items. Several comparisons are exploratory; no familywise discovery claim is made.

No true movie-dependency graph is available, so SHD/SID and edge-recovery precision are not meaningful target KPIs here. We report graph size/density, predictive error, probability quality and energy instead. A small known Gaussian model separately permits exact inference error comparisons.

Execution uses sequential CPU kernels, numerical threads set to one, and JIT warm-up before timing. Software versions, configurations, split hashes, source hashes, full logs and raw result tables are supplied. MAP/SGD fit times include validation checkpointing and scale fitting; EP fit times exclude validation scoring. Graph construction and complete search cost are reported separately. Thus wall times describe this implementation and machine, not an algorithmic speed theorem. Regenerating all runs and figures is documented in the Code README.

8 M3 results and interpretation

8.1 Prediction: several approaches under common splits

The fresh ADF/EP5 scores for seeds 42–44 reproduce the M2 values; differences in the new aggregate arise from adding seeds 45 and 46, not from replacing the old dataset with the synthetic demo. ADF/EP1 have identical natural parameters. Repeated inference improves mean RMSE from 0.8977 to 0.8643 at EP5 and 0.8548 at EP10. This supports revisiting

Method	RMSE	MAE	NLL
Global	1.1164 ± 0.0021	0.9334 ± 0.0018	1.4552 ± 0.0015
User mean	1.0342 ± 0.0021	0.8303 ± 0.0020	1.3461 ± 0.0016
Item mean	0.9786 ± 0.0026	0.7826 ± 0.0021	1.3209 ± 0.0028
MF-SGD	0.8683 ± 0.0024	0.6906 ± 0.0023	1.2214 ± 0.0025
ADF / EP1	0.8977 ± 0.0035	0.7098 ± 0.0030	1.2519 ± 0.0033
EP3	0.8740 ± 0.0036	0.6894 ± 0.0026	1.2283 ± 0.0033
EP5	0.8643 ± 0.0037	0.6808 ± 0.0025	1.2189 ± 0.0035
EP10	0.8548 ± 0.0029	0.6721 ± 0.0018	1.2098 ± 0.0024
MF-MAP (no graph)	0.8506 ± 0.0021	0.6753 ± 0.0017	1.2012 ± 0.0021
GMRF-MAP	0.8499 ± 0.0023	0.6752 ± 0.0019	1.2007 ± 0.0024

Table 4: Fresh full MovieLens 1M results, mean ± sample SD over seeds 42–46. Every row uses the expected star rating. Lower is better.

evidence with cavity removal, at increased computational cost. Ten passes remain a finite budget, not a proof of convergence.

GMRF-MAP reaches 0.8499, compared with 0.8643 for EP5 and 0.8548 for EP10. The corresponding reductions are about 1.67% and 0.57%. These are *whole-system* comparisons: Gaussian numeric-rating training, MAP fitting and regularization differ from ordinal EP. The stronger graph-free MF-MAP baseline already reaches 0.8506. Therefore most of the gain over EP5 cannot be attributed solely to adding a network.

Conventional MF-SGD remains a useful inexpensive baseline. Its expected-rating RMSE is 0.8683, while its separately decoded clipped score gives 0.8650. For MF-MAP and GMRF-MAP, clipped-score RMSE is 0.8478 and 0.8465, respectively. Those numbers are supplied as secondary decoding results; the main table uses one common expected-rating definition. Mean-only models have larger error, supporting the value of personalization on this random-interaction benchmark.

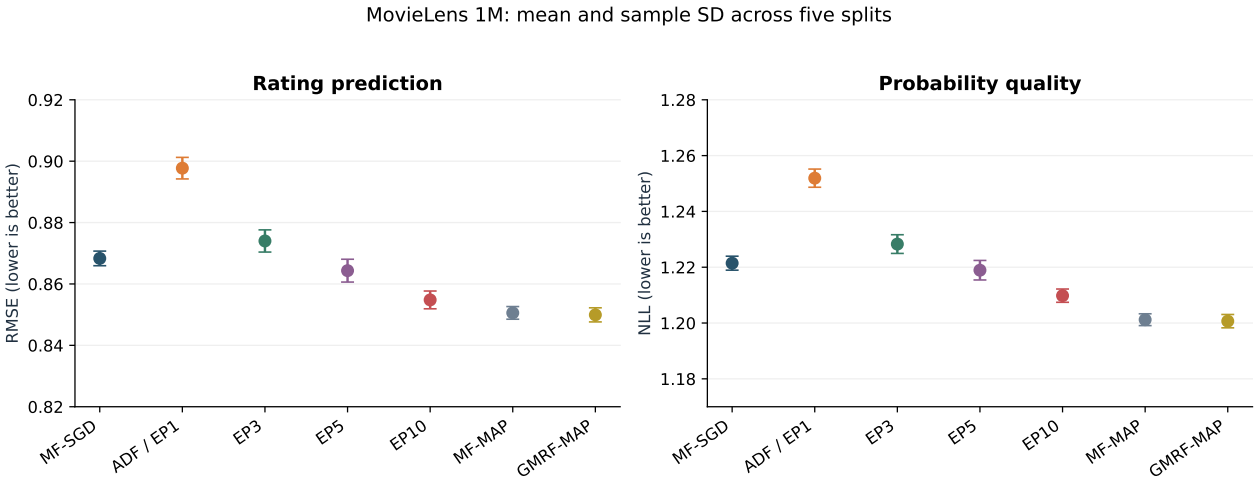


Figure 3: Accuracy and probability quality. Error bars are sample SD over five splits, not confidence intervals. The vertical scales are restricted for readability.

8.2 What improvement is actually due to the Markov prior?

All five clean-validation searches select $\gamma = 10$ over 0, 1 and 100. Against the matching $\gamma = 0$ model, the graph reduces mean expected-rating RMSE by 0.000644 (approximately 0.076%). The point estimate improves in every split, but the effect is small in practical star units.

Seed	RMSE reduction	95% user-bootstrap interval
42	+0.000941	[+0.000562, +0.001330]
43	+0.000380	[-0.000044, +0.000833]
44	+0.000492	[+0.000061, +0.000890]
45	+0.000503	[+0.000070, +0.000937]
46	+0.000906	[+0.000148, +0.001726]

Table 5: MF-MAP minus GMRF-MAP expected-rating RMSE. Positive means the graph helps. Intervals use paired user resampling within each fitted split.

Four split-specific intervals lie above zero; seed 43 includes zero. The conditional bootstrap therefore provides support in several splits without establishing a universal or independent-replication significance claim. The NLL difference is also small, and its interval includes zero for several splits (full table in the results files). We do not claim improved calibration from the graph: mean ECE changes from 0.03835 to 0.03868, slightly worse.

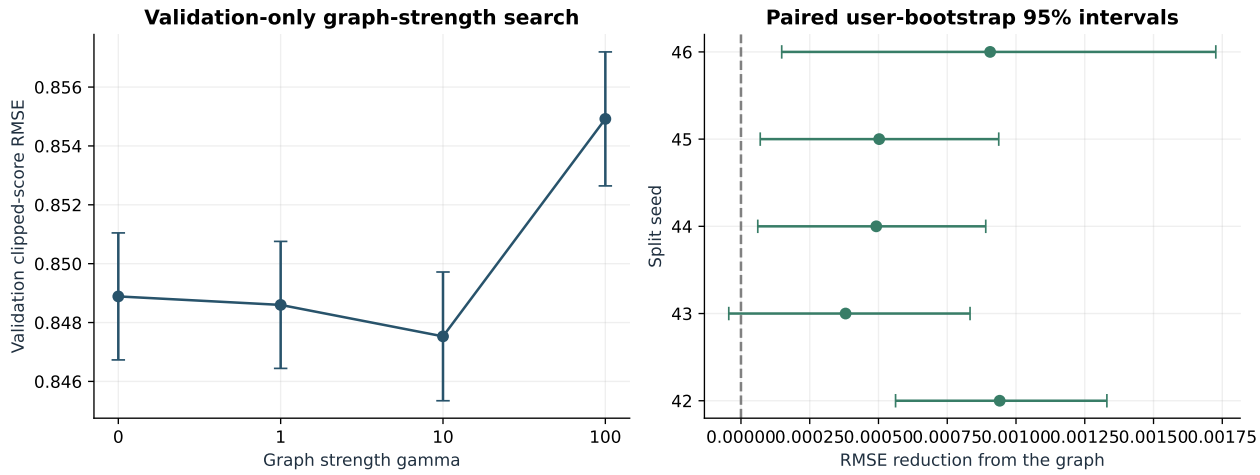


Figure 4: Graph-strength validation and the paired graph-only test effect. Strength 100 worsens validation error; stronger smoothing is not always better.

The mechanism is consistent with modest regularization: neighbouring movie vectors share evidence while the data term retains personalization. The ablation establishes the effect of this prior *within this solver and selection protocol*; it does not prove that the selected neighbours are true dependencies or explain each gain causally. The stress tests below examine where this explanation remains useful and where it fails.

8.3 Probability quality and the two noise questions

More EP passes improve RMSE, NLL and Brier, but worsen ECE: 0.0451 for ADF, 0.0566 for EP5 and 0.0608 for EP10. Thus lower rating error does not mean more calibrated confidence. A global empirical distribution has low ECE but poor personalized error, illustrating that calibration alone is insufficient. GMRF-MAP has better NLL than EP5/EP10 here, while its

Method	Brier	ECE	Star accuracy
MF-SGD	0.6554 ± 0.0011	0.0397 ± 0.0010	46.04 ± 0.16%
ADF / EP1	0.6681 ± 0.0012	0.0451 ± 0.0010	44.13 ± 0.13%
EP3	0.6590 ± 0.0012	0.0537 ± 0.0017	45.20 ± 0.18%
EP5	0.6552 ± 0.0012	0.0566 ± 0.0009	45.59 ± 0.16%
EP10	0.6516 ± 0.0007	0.0608 ± 0.0012	46.04 ± 0.21%
MF-MAP (no graph)	0.6467 ± 0.0007	0.0383 ± 0.0018	46.98 ± 0.15%
GMRF-MAP	0.6468 ± 0.0009	0.0387 ± 0.0022	46.96 ± 0.19%

Table 6: Probability quality and exact-star accuracy (percent), five-seed mean ± SD. Accuracy uses the most probable star, not rounded expected stars.

probability model still omits posterior parameter uncertainty. Nominal 90% equal-tail intervals cover 98.22% of ratings for EP5 and 97.41% for GMRF-MAP, with mean widths 3.82 and 3.60 star categories. Such broad, discrete intervals should not be advertised as well calibrated merely because their coverage is high.

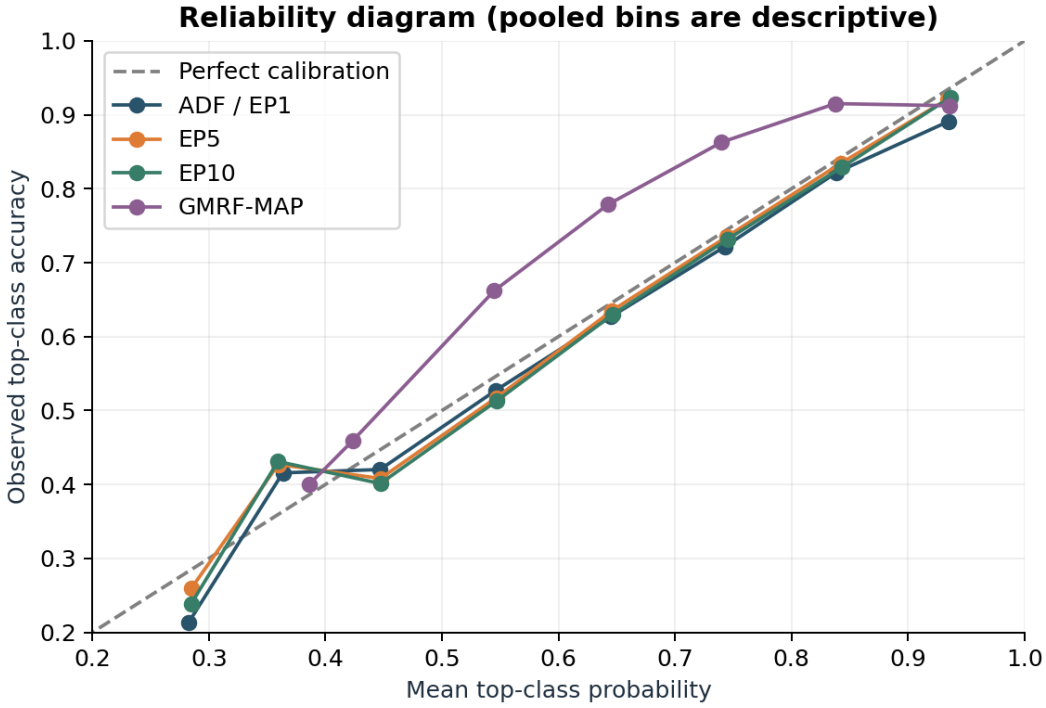


Figure 5: Top-class reliability. Pooling overlapping split bins is descriptive; the curve is not an independent-data significance analysis.

Noise variance	RMSE	NLL	ECE
0.5	0.8691 ± 0.0036	1.2404 ± 0.0054	0.0339 ± 0.0012
1.0	0.8643 ± 0.0037	1.2189 ± 0.0035	0.0566 ± 0.0009
2.0	0.8743 ± 0.0030	1.3154 ± 0.0020	0.0856 ± 0.0012

Table 7: EP5 observation-noise variance sensitivity. This changes the assumed likelihood variance, not the dataset's labels.

Validation NLL chooses $\beta^2 = 1$ in every seed. Halving it improves ECE to 0.0339 but worsens NLL to 1.2404 and RMSE to 0.8691. Doubling it worsens all three displayed measures. A

smaller assumed variance sharpens probabilities; the data do not support a blanket claim that reducing noise always improves inference. The separate label-corruption experiment measures resistance to wrong training observations while keeping evaluation labels unchanged.

8.4 Computational cost, convergence and exact-reference checks

Method	Fit (s)	Predict (μs/rating)	Site MiB
MF-SGD	1.38 ± 0.04	0.413	0
ADF / EP1	3.33 ± 0.19	0.810	0
EP3	10.99 ± 0.57	0.749	219.77
EP5	18.29 ± 2.02	0.679	219.77
EP10	35.99 ± 6.61	0.779	219.77
MF-MAP (no graph)	16.74 ± 2.74	0.449	0
GMRF-MAP	14.48 ± 1.82	0.487	0

Table 8: Selected-candidate fitting and batch prediction cost; mean ± SD where shown. EP site memory is an array-size calculation, not process RSS.

EP5 takes about 5.49 times ADF's mean fit time; EP10 about 10.81 times. Both repeated-EP budgets use the same 219.77 MiB cache at $K = 8$; more passes increase time rather than site count. MF-SGD fits in about 1.38 seconds under this implementation. Mean graph construction takes 4.31 seconds and the full four-candidate MAP search takes 66.37 seconds, beyond the selected GMRF candidate's 14.48 seconds. A claim that GMRF is simply "faster than EP" would ignore this search/setup cost. Selected MAP wall times vary despite equal epoch budgets; they do not establish an intrinsic speed advantage from graph edges.

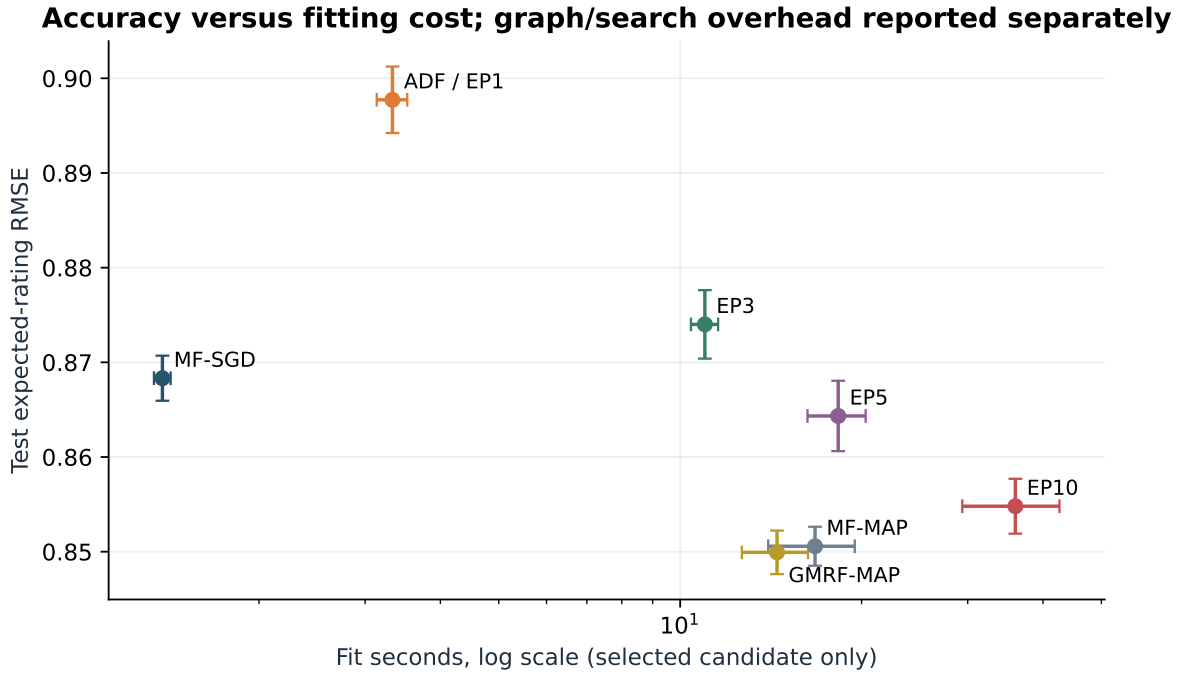


Figure 6: Prediction error versus fitting time. Points omit graph building and search overhead, which are stated explicitly in the text and raw results.

At EP pass 10, mean RMS parameter change remains 0.01564, despite no skipped updates in the main EP traces. The local cap is reached by an average of 8.8 updates out of 800,167

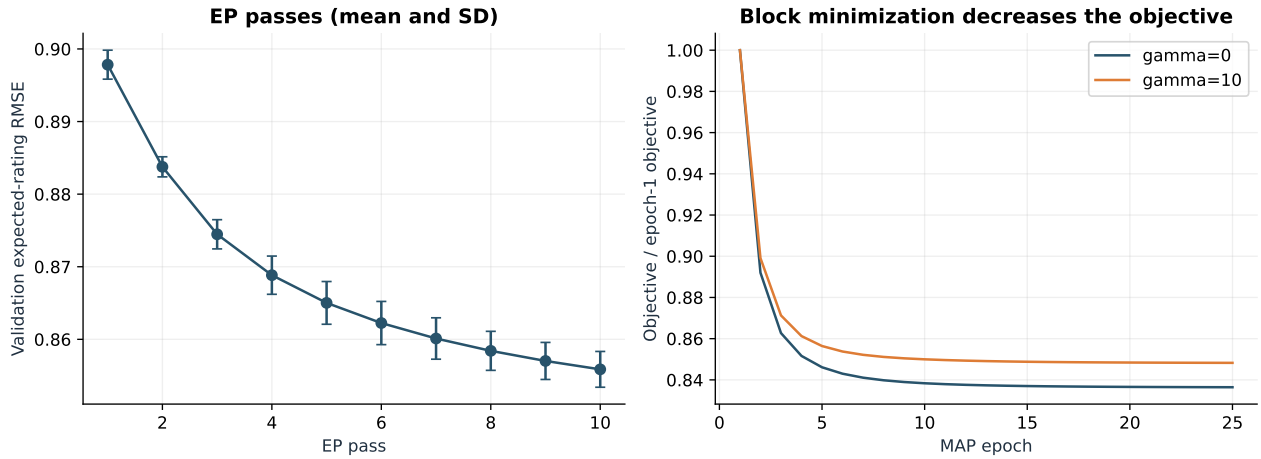


Figure 7: Validation error by EP pass and relative MAP objective by epoch. Lower energy does not guarantee lower held-out error, so validation selects checkpoints. EP is not claimed to optimize this MAP energy.

at that pass. Stable local calculations therefore do not establish that the outer posterior approximation has converged.

A separate 12-variable Gaussian Markov diagnostic compares a dense exact Gaussian solve, Gauss–Seidel MAP iterations and Gibbs sampling. The diagnostic uses a 12-node chain with an extra edge of weight 0.5 between zero-based nodes 2 and 8, $J = 0.8I + 2L + \mathrm{diag}(d)$ with d evenly spaced from 0.2 to 1.5, and $h_i = \sin(i + 0.4)$. For $p(q) \propto \exp(-q^T Jq/2 + h^T q)$, the known posterior has mean $J^{-1}h$ and covariance J^{-1}. Gauss–Seidel approaches the exact mean; Gibbs estimates both means and variances with Monte Carlo error. Increasing sample budget reduces error on average but not necessarily for every seed. These are additional inference approaches on a tractable diagnostic, **not full MovieLens recommender scores**.

Method	Budget	Mean RMSE	Variance RMSE	Time (ms)
ExactGaussian	1	0	0	1.258
GaussSeidelMAP	1	0.138	–	0.018
GaussSeidelMAP	5	0.00251	–	0.007
GaussSeidelMAP	20	4.46e-08	–	0.014
GaussSeidelMAP	100	2.17e-17	–	0.066
Gibbs	1000	0.0185	0.0134	2.652
Gibbs	5000	0.0111	0.00689	8.790
Gibbs	20000	0.00383	0.00333	36.379

Table 9: Small Gaussian diagnostic. Gibbs values average five chains/seeds; 1,000 burn-in sweeps are excluded from the sample count but included in time. MAP has no variance estimate, hence "–". Exact/iterative timings are single small-system measurements. Full variation is in the CSV.

The original bilinear ordinal quadrature diagnostic is also rerun. It exposes the same variance/covariance limitations of the factorized approximation; success on the Gaussian diagnostic does not make bilinear ordinal EP exact. For a dense n-variable Gaussian model, direct solution costs $O(n^3)$, while a dense iterative or Gibbs sweep costs $O(n^2)$; sparse neighbours can reduce sweep work. Those guarantees do not turn the full bilinear ordinal posterior into a Gaussian linear system.

9 Robustness, ablations and failure analysis

9.1 Missing ratings and corrupted observations

Condition	ADF	EP5	MF-MAP	GMRF-MAP
main	0.8980 ± 0.0049	0.8633 ± 0.0040	0.8507 ± 0.0028	0.8501 ± 0.0031
retain_0.25	0.9217 ± 0.0041	0.9199 ± 0.0051	0.9178 ± 0.0042	0.9100 ± 0.0038
retain_0.5	0.9102 ± 0.0039	0.8920 ± 0.0049	0.8810 ± 0.0039	0.8775 ± 0.0041
noise_0.1	0.9129 ± 0.0036	0.8910 ± 0.0036	0.8827 ± 0.0029	0.8797 ± 0.0030
noise_0.2	0.9386 ± 0.0030	0.9326 ± 0.0022	0.9317 ± 0.0023	0.9248 ± 0.0018
cold_items	1.0703 ± 0.0079	1.0781 ± 0.0068	1.0622 ± 0.0060	1.0545 ± 0.0077
cold_users	0.9837 ± 0.0137	0.9861 ± 0.0146	0.9935 ± 0.0115	0.9916 ± 0.0117

Table 10: Expected-rating test RMSE, mean ± sample SD over seeds 42–44. The clean row uses these same three seeds. Cold-start rows evaluate different restricted test populations and should not be ranked directly against the clean row.

Reducing training availability increases error for all methods. At 25% retention, the graph reduces MF-MAP RMSE from 0.9178 to 0.9100; at 50%, from 0.8810 to 0.8775. The corresponding gains, 0.0077 and 0.0036, are larger than the clean three-seed gain of 0.0006. Sharing evidence is more useful when each movie has less direct support. This supports the variance-reduction hypothesis under random removal, while not establishing robustness to actual nonrandom exposure or missing-not-at-random ratings.

With 10% corrupted labels, MF-MAP and GMRF-MAP have RMSE 0.8827 and 0.8797; at 20%, 0.9317 and 0.9248. The graph mitigates some degradation but does not recover clean-data performance. Both the training target and the residual similarities are affected by corruption. The graph helps here despite that imperfect evidence, consistent with useful smoothing; this experiment does not identify a universally optimal graph or realistic user-error mechanism.

EP5 remains better than ADF in aggregate, but its advantage becomes small under sparse or heavily corrupted training. At 25% retention, seed 43's EP5 is slightly worse than ADF. Repeated inference cannot create reliable evidence that is absent and may reinforce misspecified observations. These stress comparisons are descriptive three-seed results, not independent-dataset significance tests.

9.2 Which graph components matter?

Graph condition	MF-MAP RMSE	GMRF-MAP RMSE
main	0.8507 ± 0.0028	0.8501 ± 0.0031
ratings_only	0.8507 ± 0.0028	0.8499 ± 0.0031
genres_only	0.8507 ± 0.0028	0.8510 ± 0.0031
graph_top_5	0.8507 ± 0.0028	0.8497 ± 0.0031
graph_top_50	0.8507 ± 0.0028	0.8505 ± 0.0031

Table 11: Graph ablations at fixed $\gamma = 10$, three seeds. The graph-free baseline is unchanged across clean graph variants.

Ratings-only edges give mean RMSE 0.8499, mixed edges 0.8501, and genre-only edges 0.8510, versus 0.8507 without a graph. Thus coarse genres alone do not improve this warm-interaction

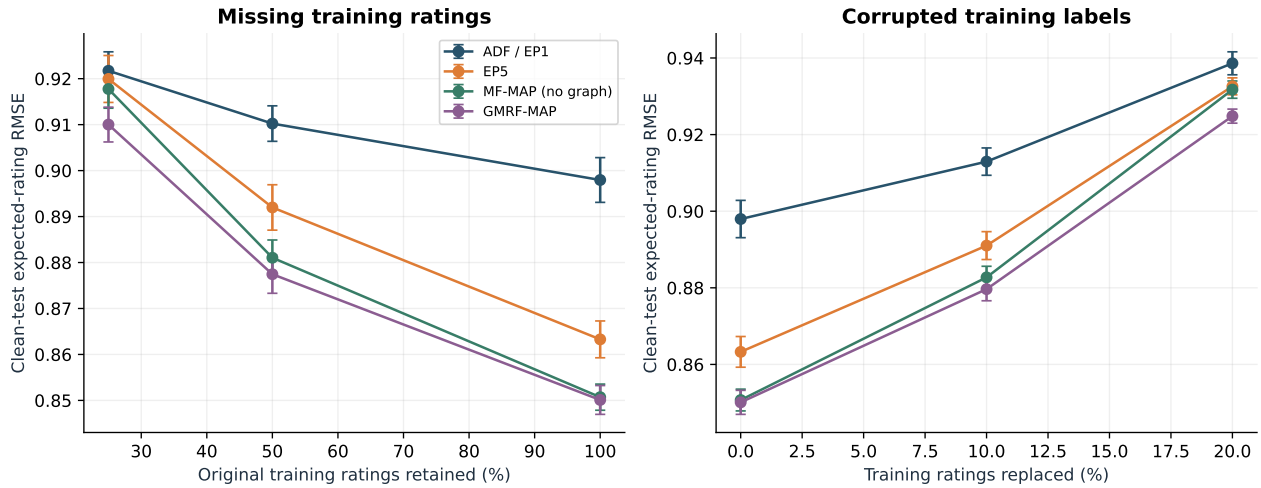


Figure 8: Missing-data and label-noise sensitivity. All evaluations use clean test labels; error bars show sample SD across the same three seeds.

benchmark. They can connect movies whose audiences or rating patterns differ; adding metadata is not automatically beneficial. Genre information still supplies connections for unseen items, which is a different evaluation setting.

Top-5 neighbours give RMSE 0.8497, whereas top-50 gives 0.8505. More edges do not necessarily carry more useful information. Combined with the poorer validation result at strength 100, this reveals an oversmoothing risk. These topology variants are exploratory diagnostics. The main configuration remains top-20, fixed before these test comparisons; we do not select a new reported winner using their test scores.

9.3 Controlled cold-start and remaining errors

For withheld movies, GMRF-MAP improves the matching baseline from 1.0622 to 1.0545 RMSE. Genre connections can transfer neighbouring traits when a movie has no training observations, but its own rating bias cannot be estimated. Error remains much higher than on the warm main benchmark. EP5 is worse than ADF here (1.0781 versus 1.0703), another counterexample to universal EP superiority.

For withheld users, ADF has RMSE 0.9837, EP5 0.9861, MF-MAP 0.9935 and GMRF-MAP 0.9916. The movie graph offers no user metadata or user-specific evidence, so it cannot personalize a completely unseen user. It slightly helps its matching MAP baseline but does not beat ADF in this setting. This limits the claim that a movie network resolves cold-start in general.

The main benchmark's five-star error is 1.0245 for GMRF-MAP versus 0.9748 for EP10, despite GMRF's better aggregate RMSE. Its advantages are stronger around three/four-star observations. The expected-rating decision and regularization shrink predictions toward the centre, overpredicting low ratings and underpredicting high ratings. An overall score therefore hides meaningful class-specific failures.

Low-support movies are harder for every method. The natural zero-training-item bin contains only 83 pooled test observations across five overlapping splits; its curve should not support a broad cold-start claim. The controlled withheld-item experiment is the more appropriate evidence. Support strata also differ in movie composition, so their error differences are descriptive, not causal estimates of adding ratings.

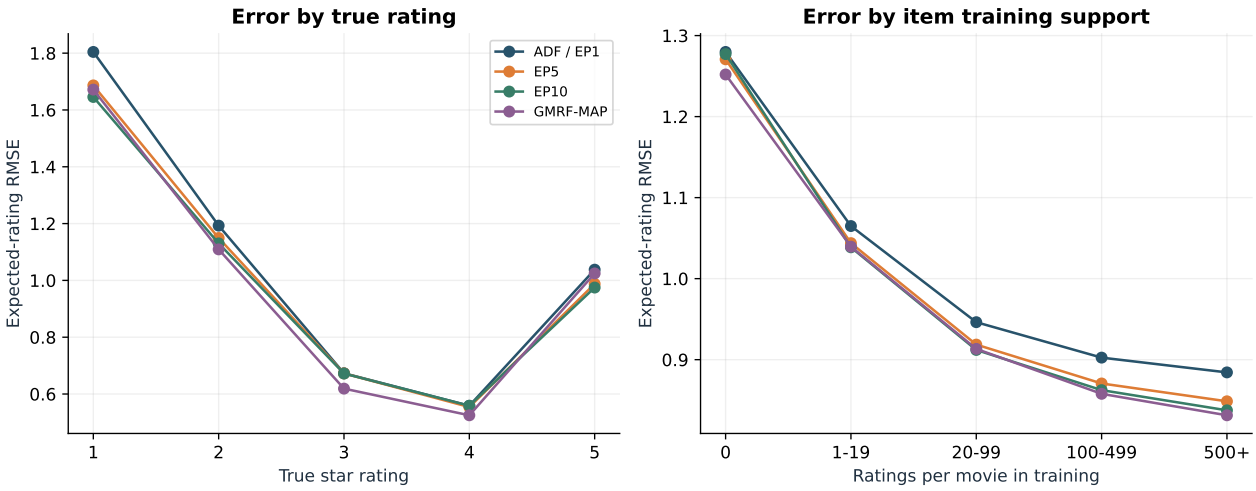


Figure 9: Descriptive pooled errors by true rating and movie support. Pooled split observations overlap; the curves do not represent independent samples.

9.4 Prior, damping and order sensitivity

EP5 variant	RMSE	NLL
ADF_reshuffled	0.8984 ± 0.0049	1.2526 ± 0.0046
damping_0.25	0.8752 ± 0.0048	1.2292 ± 0.0041
damping_1	0.8529 ± 0.0038	1.2078 ± 0.0027
prior_0.1	0.8672 ± 0.0043	1.2216 ± 0.0036
prior_0.5	0.8650 ± 0.0039	1.2195 ± 0.0031
reshuffled	0.8638 ± 0.0039	1.2183 ± 0.0030

Table 12: Fixed EP5 sensitivity variants, mean ± SD over seeds 42–44. Each varies one operating choice; the main configuration is not reselected using these test outcomes. Reshuffled ADF is the first pass of that order.

Prior strength changes regularization, damping changes the size of cached-site replacements, and reshuffling changes the sequence of approximate updates. These comparisons expose dependence on operating choices rather than proving that the original settings are optimal. Their validation trajectories and test metrics are supplied together. A broader validation-only search would be needed to compare optimized method families fairly; this finite study does not claim exhaustive tuning or order invariance.

With undamped replacement after the first pass (outer damping 1), EP5 reaches mean RMSE 0.8529 across the three sensitivity seeds, compared with 0.8633 for the original EP5 setting and 0.8501 for GMRF-MAP on those same seeds. Thus tuning the inference schedule closes much of the apparent gap. This is an exploratory fixed-variant comparison, not a validation-selected replacement for the five-seed main method. It further limits any claim that graph structure alone explains superior performance.

10 Related work and revised SOTA positioning

Matchbox: Microsoft Research, WWW 2009. Stern, Herbrich and Graepel [4] combine latent features, optional metadata, feedback models and hybrid message passing. ADF

provides efficient online Bayesian updates; repeated EP revisits cached rating contributions. Their experiments study recommendation accuracy, cold-start behaviour and scalability. Our implementation retains the inference question but uses static ID features, fixed cuts and its own schedule, rather than reproducing Infer.NET or the full system. Published scores use different protocols and are not directly comparable.

PreRec: AWS AI Labs affiliation, WSDM 2024. Lin et al. [3] use a hierarchical Bayesian model with neural user/item encoders, domain variables and popularity effects. Training uses MAP optimization; zero-shot scoring changes domain and popularity mechanisms. XMarket/Online Retail experiments evaluate cross-domain sequential ranking and limited-data adaptation with Recall and revised NDCG metrics. Reported gains support modelling transfer and bias in that setting, subject to the learned representations and assumed bias mechanisms. They do not establish an EP advantage or provide a MovieLens rating baseline.

The evidence revises our position from an implementation comparison to a controlled inference-and-regularization study. Longer EP passes improve prediction at extra time and imperfect calibration. A GMRF adds a small benefit over an already strong MAP baseline; robustness experiments limit any claim of general superiority. The implemented contributions are comparative experiments, mathematical checks and interpretation, not a new inference algorithm or causal-debiasing system. Neither published paper's scores are numerically compared with ours because the evaluation protocols differ.

11 Limitations and conclusion

The model is static and evaluated through random interactions, not future recommendations. Learned latent coordinates are not uniquely identifiable semantic traits. EP retains a factorized approximation and cannot represent all posterior correlations. Its moment and convergence checks validate specific properties rather than global exactness. GMRF-MAP uses point estimates, and its fitted residual scale does not recover Bayesian parameter uncertainty. Similarity edges can be wrong, particularly under sparse or corrupted observations; neither genres nor correlations establish causality.

The parameter searches are finite and not an exhaustive optimum comparison. Several selected checkpoints reach the budget boundary. SGD and MAP have different regularization conventions, and the EP/MAP likelihoods differ. Changing them jointly is useful for a system comparison but does not isolate the graph. Confidence intervals are conditional on fitted models and resample users only; overlapping splits and shared-item dependence restrict the scope of significance claims. Missingness/corruption are controlled artificial stress mechanisms, not a full model of actual user reporting or exposure bias.

M3 answers the M2 feedback with multiple fitting/inference approaches, a mathematically specified Markov network, matched ablations, explicit noise experiments, probability-quality measures and statistical interpretation. On the main benchmark, GMRF-MAP's RMSE is 0.8499 ± 0.0023, versus 0.8643 ± 0.0037 for EP5 and 0.8548 ± 0.0029 for EP10. The comparison that isolates the graph is much closer: 0.8506 ± 0.0021 without it versus 0.8499 ± 0.0023 with it. The observed graph gain is therefore modest, with split-dependent statistical support, while its benefit is more visible in some stress conditions.

The defensible explanation is that the pairwise prior can share evidence between related movies and reduce estimation variance, while excessive or unhelpful connections introduce bias. This mechanism and the matched ablation support a qualified improvement claim; they do not imply that any Markov or Bayesian network will improve any recommender. Repeated EP likewise

improves prediction without uniformly improving calibration. The project's contribution is the reproducible evidence about these trade-offs and limitations.

Next steps are learned user-specific thresholds, chronological evaluation, better treatment of nonrandom missingness, broader validation-only tuning, and a posterior approximation that retains graph-related covariance. These are proposed extensions, not completed experiments or claims in this submission.

References

[1] GroupLens Research. MovieLens 1M Dataset. https://grouplens.org/datasets/movielens/1m/. Official archive downloaded and SHA-256 verified on 4 November 2026.

[2] F. Maxwell Harper and Joseph A. Konstan. The movielens datasets: History and context. *ACM Transactions on Interactive Intelligent Systems*, 5(4), 2015. doi:10.1145/2827872.

[3] Ziqian Lin, Hao Ding, Nghia Trong Hoang, Branislav Kveton, Anoop Deoras, and Hao Wang. Pre-trained recommender systems: A causal debiasing perspective. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, pages 424–433, 2024. doi:10.1145/3616855.3635779.

[4] David H. Stern, Ralf Herbrich, and Thore Graepel. Matchbox: Large scale online bayesian recommendations. In *Proceedings of the 18th International Conference on World Wide Web*, pages 111–120, 2009. doi:10.1145/1526709.1526725.

A Requirement coverage and report continuity

The six M3 notice screenshots, the general CSE 516 project PDF and the L5 guidelines slides were reviewed. The M3 notice is more specific than the older general requirement: M3 requires the updated PDF and root README; code/results are optional, and there is no video requirement. The notice gives 4 November 2026, 11:59 PM as the deadline. This package does not assert that submission has already occurred. Each member submits the individual form separately.

M3 expectation	Evidence in the cumulative report/package
Full experimental results	Five main seeds; three stress-test seeds; raw metrics and configs
Baseline comparison	M2 means/SGD/ADF/EP; graph-free MAP and matching GMRF
KPI plots and tables	Rating/probability errors, cost, reliability, convergence, stress and errors
Variation and uncertainty	Sample SD; 2,000-replicate paired user-bootstrap intervals with limits
Probability quality	NLL, Brier, ECE, reliability and interval outputs
Robustness	Training availability, label corruption, assumed noise, graph density, cold items/users
Ablation and error analysis	Gamma=0, genre/residual components, rating/support groups
Interpretation	Main-result explanations, counterexamples and explicit attribution limits
Theoretical validation	Preserved EP derivation; proper GMRF prior; exact/Gibbs/iterative diagnostic; quadrature
Revised SOTA position	Specific Matchbox/PreRec models and protocols; no direct SOTA score claim
Reproducibility	Source/config/environment hashes, run logs, tests, result/figure scripts
Individual contribution	Inherited roles marked historical; actual M3 contributions require confirmation

Table 13: Coverage of the supplied guidelines. SHD is explicitly inapplicable without a true graph.

The updated source begins from the revised M2 `main.tex` and preserves its title, team details, model equations, EP/VMP derivation and numerical-validation discussion. It updates the abstract, milestone/date, benchmark, results, positioning and conclusion; it adds the GMRF representation and M3 analysis. Detailed changes and code locations are supplied in the Review folder.

Not every item mentioned as possible future work in M2 is claimed complete. The implemented additions cover noise/prior/damping/order sensitivity, longer EP runs, genre ablations, reduced data and cold-start evaluations. Learned user-specific thresholds, a temporal deployment test, and full-posterior GMRF inference remain deferred. The formal general guidelines encourage depth in one question, rather than an unsupported claim to have tested every PGM method.

B Team accountability and AI assistance

The historical role plan is retained: Jyotier Vithalani (AU2440170): mathematical model, EP/ADF and repository; Tanmay Kanani (AU2440234): baselines; Ansh Chaudhari (AU2440259): paper/SOTA analysis; Shlok Shah (AU2440178): dataset and metadata; Het Patel (AU2440221): KPIs, experiment tracking and documentation. These are inherited assignments, not verified records of who completed M3. Each member must identify actual work in the individual form and be able to explain the entire pipeline.

Generative AI assisted source preparation, debugging, execution, analysis, figures and writing. New numerical results were computed by the supplied code on the verified dataset; they were not fabricated to make a PGM win. The students remain responsible for checking and understanding every claim. The Review folder includes a viva guide and concrete responsibilities for verification, without certifying that the team has already performed them.

C Reproduction and evidence boundaries

The submission folder is `M3_g4_th_apl_recommend-sys_team`, intended for the same existing repository used for M2. No new repository is required. The Code README provides commands for validation, main/stress/sensitivity experiments, figure regeneration and report compilation. Raw MovieLens files are not redistributed; the official downloader, source links and hashes are included. Seed-42 checkpoints and ID mappings support direct probability queries.

The main-run source snapshot and its hashes preserve the exact executed implementation. Later changes add a cold-user condition and audit metadata to the robustness path, without changing the main numerical kernels or selection rules. Results from the separate main, robustness and sensitivity stages retain their individual configurations and source/environment records.

The earlier synthetic offline demo is excluded from all MovieLens benchmark tables. The 12-variable Gaussian example is labelled a diagnostic throughout. Historical M2 files are labelled separately. Training time excludes JIT and test evaluation; comparison of complete workflows additionally includes graph construction and validation search. Test evaluation follows validation selection for each comparison, and changing a diagnostic after seeing a test result is not presented as a preregistered improvement.